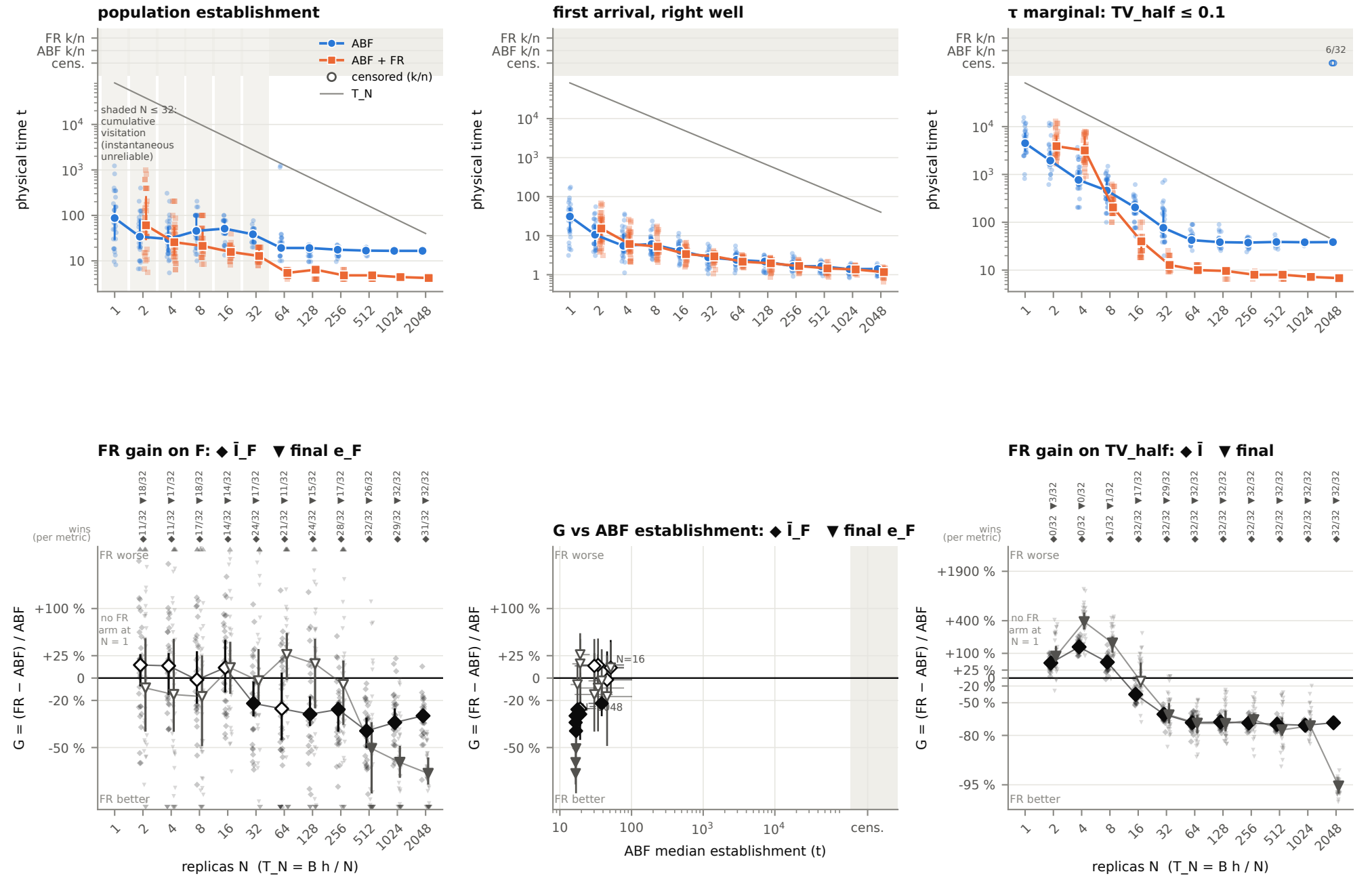


entropic gateway: population establishment, first arrival and marginal convergence vs N (physical time)

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B h / N$); 32 seeds per (N, method). Do FR gains coincide with establishment-limited N?



entropic gateway · S6b_establishment_time · status: all planned (N, seed, method) complete · FR is not defined at N = 1 (ABF only, by design).
Establishment = first save after which the instantaneous far-state fraction (right well $x > 0.5$) stays $\geq \frac{1}{2} \times 0.361$. Shaded N [1, 2, 4, 8, 16, 32]: the instantaneous rule is UNRELIABLE there (ill-defined at $N \leq 2$; an exactly uniform population passes every save of the second half with probability < 0.9 , summary establishment_null), so the same rule on the cumulative visitation is plotted; FR birth-death can push the instantaneous count below binomial scatter, so an FR establishment advantage at those N is not evidence of faster convergence. $\tau(TV_half)$ on the budget axis uses the common budget grid.
Top: seed median (large open marker in the 'cens.' band = median censored), IQR bars, faint seeds. Bottom: paired G with bootstrap 95 % CIs (filled = excludes 0; 'wins' = $FR < ABF / n$); scatters: each N's G against the ABF median time (horizontal bar = ABF IQR; 'cens.' = censored).
Source results/equal_budget_v2/gateway/analysis/summary.json (generated 2026-10-10T08:08:27Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.